

Lab sheet 19**Submission Due: 30.10.2024****Title:** React Routes and API**Aims:**

- Creating a dynamic list
- Using context to share data between components
- Building a simple routing system
- API and Axios

Tasks:

- Use **useState** to manage list
- Use **Context API** to pass data between components without props drilling.
- Use **React Router** for client-side routing
- Use **Axios** to fetch data from an API

Activities :

1. Use **useState** to manage list

```
import { useState } from 'react';

const DynamicList = () => {
  const [items, setItems] = useState([]);
  const addItem = () => setItems([...items, `Item ${items.length + 1}`]);

  return (
    <div>
      <button onClick={addItem}>Add Item</button>
      <ul>
        {items.map((item, index) => <li key={index}>{item}</li>)}
      </ul>
    </div>
  );
};

export default DynamicList;
```

2. Use Context API to pass data between components without props drilling.
 - a. ContextProvider.jsx

```
import { createContext, useState } from 'react';

export const MyContext = createContext();

const ContextProvider = ({ children }) => {
  const [sharedData, setSharedData] = useState("Initial Data");

  return (
    <MyContext.Provider value={{ sharedData, setSharedData }}>
      {children}
    </MyContext.Provider>
  );
};

export default ContextProvider;
```

- b. ComponentNew.jsx

```
import { useContext } from 'react';
import { MyContext } from './ContextProvider';

const ComponentNew = () => {
  const { sharedData, setSharedData } = useContext(MyContext);

  return (
    <div>
      <p>{sharedData}</p>
      <button onClick={() => setSharedData("Updated Data")}>Update</button>
    </div>
  );
};

export default ComponentNew;
```

3. Use React Router for client-side routing

Install React Router :

npm install react-router-dom

- a. Home.jsx

```
const Home = () => {
  return <h2>Home</h2>;
};
```

```
export default Home;
```

b. About.jsx

```
const About = () => {  
  return <h2>About</h2>;  
};  
  
export default About;
```

c. AppRoutes.jsx

```
import { Routes, Route } from 'react-router-dom';  
import Home from './Home';  
import About from './About';  
  
const AppRoutes = () => {  
  return (  
    <Routes>  
      <Route path="/" element={<Home />} />  
      <Route path="/about" element={<About />} />  
    </Routes>  
  );  
};  
  
export default AppRoutes;
```

d. App.jsx

```
import { BrowserRouter as Router, Link } from 'react-router-dom';  
import AppRoutes from './AppRoutes';  
  
const App = () => (  
  <Router>  
    <nav>  
      <Link to="/">Home</Link>  
      <Link to="/about">About</Link>  
    </nav>  
    <AppRoutes />  
  </Router>  
  
export default App;
```

4. Use Axios to fetch data from an API

Install Axios :

npm install axios

a. DataFetchingAndPosting.jsx

```
import { useEffect, useState } from 'react';
import axios from 'axios';

const DataFetchingAndPosting = () => {
  const [data, setData] = useState([]);
  const [newPost, setNewPost] = useState({
    title: '',
    body: '',
    userId: 1
  });

  // GET request to fetch data
  useEffect(() => {
    axios.get('https://jsonplaceholder.typicode.com/posts')
      .then(response => setData(response.data))
      .catch(error => console.error(error));
  }, []);

  // POST request to add new data
  const handlePost = () => {
    axios.post('https://jsonplaceholder.typicode.com/posts',
newPost)
      .then(response => {
        console.log('New Post Added:', response.data);
        setData([...data, response.data]);
      })
      .catch(error => console.error(error));
  };

  return (
    <div>
      <h3>Data Fetching (GET Request)</h3>
      <ul>
        {data.map(post => <li key={post.id}>{post.title}</li>)}
      </ul>

      <h3>Add a New Post (POST Request)</h3>
      <input
        type="text"
        placeholder="Title"
        value={newPost.title}
        onChange={e => setNewPost({ ...newPost, title:
e.target.value })}
      />
      <input
        type="text"
        placeholder="Body"
```

```
        value={newPost.body}
        onChange={e => setNewPost({ ...newPost, body:
e.target.value })}
      />
      <button onClick={handlePost}>Add Post</button>
    </div>
  );
};

export default DataFetchingAndPosting;
```

b. App.jsx

```
import DataFetchingAndPosting from './DataFetchingAndPosting';

function App() {
  return (
    <>
      <DataFetchingAndPosting />
    </>
  );
}

export default App;
```

Discussion:

- How does Axios differ from *fetch*?
- What happens when the component mounts with *useEffect*?